# Badr Albdulmajeed

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## ABOUT ME

I'm Badr Albdulmajeed, a final-year Electrical and Electronic Engineering student at Asia Pacific University. I'm passionate about applying my engineering knowledge to build practical, working solutions and enjoy seeing how different fields come together, from circuit design, power systems and control systems to embedded AI and software. My recent internship and university projects allowed me to integrate these areas by developing automation systems using hardware, PLCs, machine vision, power system analysis and real-time GUI tools. I enjoy working in a team to solve challenges and am ready to apply my engineering skills to a new project.

## SKILLS AND STRENGTHS

- Technical Skills:
    - Microprocessors & Embedded Systems: Architecture, programming and interfacing using Raspberry Pi 5, NVIDIA Jetson Orin Nano, ESP32, Arduino and sensors.
    - Computer Vision & AI: OpenCV, Roboflow, YOLOv8, Vision Transformer (ViT), PyTorch, ONNX Runtime GPU, object detection, segmentation, image classification, data annotation and augmentation.
    - Robotics & Automation: AGV vision system development, ZED 2i stereo camera, RGB-D depth sensing, LiDAR data integration, RFID access control and real-time GUI testing.
    - Power Systems & Protection: Power flow analysis, overcurrent relay coordination, IDMT relay settings, IEC/IEEE inverse curves, TCC plotting, fault analysis, CTs and circuit breaker protection logic.
    - Simulation & Design: MATLAB, Simulink, MATLAB App Designer, LabVIEW, PowerWorld Simulator, SolidWorks, Dialux, LTspice and Tinkercad.
    - PLC and Pneumatics: Siemens PLC S7-300, S7-200, S7-1200, TIA Portal SCL, PLC-controlled automation workflow and pneumatic circuits.
    - Programming & GUI Development: Python, C, C++, C#, PyQt5, Streamlit, MATLAB GUI/App Designer and basic web/IoT dashboard work.

Programs and Software:

- MATLAB, Simulink, MATLAB App Designer, LabVIEW, PowerWorld Simulator, TIA Portal, Roboflow, OpenCV, PyTorch, Ultralytics YOLOv8, ONNX Runtime GPU, PyQt5, ZED SDK, CUDA, JetPack, Ubuntu Linux, Blynk, Arduino IDE, SolidWorks, Dialux, LTspice and Tinkercad.

- Project Management:
  - Successfully led and managed multiple university group projects, developing strong team foundation, budgeting skills, and time management.
- Soft Skills:
  - Strong work ethic, reliability, excellent communication, work efficiently as a team member and the ability to quickly adapt to new environments.

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## EXPERIENCE

**R&D Project Engineer at Control easy** [March 2025] – [July 2025]

- Developed and deployed multiple end-to-end machine vision systems for industrial quality control, applying a range of AI techniques from foundational concepts to advanced applications.
  - **Classification:** Built a system to classify different fabric types and engineered a capstone automated station using a Vision Transformer (ViT) model to classify and sort golf balls in real-time.
  - **Object Detection:** Created applications to identify patterns on fabrics and to locate and verify the presence of specific electronic components on PCBs.
  - **Segmentation:** Implemented models for precise defect analysis, including outlining scratches and holes on wood surfaces and analyzing component placement on circuit boards.
- Engineered and built the complete hardware and software for an automated inspection system, safely interfacing a 24V industrial sensor with a 3.3V Raspberry Pi 5 using a relay circuit.
- Integrated vision systems with an Industry 4.0 machine, ensuring compatibility with its existing PLC-controlled automation workflow.
- Received hands-on training with Siemens PLCs (S7-300, S7-200, S7-1200), including programming in (SCL) within the TIA Portal.
- Authored complete technical manuals for all projects, detailing the entire process from dataset creation and model training to GUI development and hardware setup.

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## EDUCATION

- University (2020 – present)
  - Asia Pacific University of Technology & Innovation (APU)
    - BACHELOR OF ELECTRICAL AND ELECTRONIC ENGINEERING WITH HONORS
    - Expected Graduation (2026)
- Sama American School (2017 – 2019)
  - American High school Diploma

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## PROJECTS

1. Automated Golf Ball Quality Control System: Engineered a complete, automated inspection station using a Raspberry Pi 5 to process signals from a proximity sensor, trigger a camera, and run a Vision Transformer (ViT) model for real-time classification.
2. PCB Component Inspection System: Developed an intelligent verification system using object detection and segmentation to locate present components and logically deduce missing ones. Implemented custom logic and computer vision heuristics to eliminate false positives.
3. Wood Defect Segmentation System: Built an application using instance segmentation to precisely find and outline defects like scratches and holes on wood surfaces, providing a quantitative quality score.
4. Fabric Quality Control Systems: Created foundational machine vision applications for fabric classification and object detection, deploying them on both PC and embedded (Raspberry Pi 5) platforms.
5. Vaccination Center System (Python): Developed a system to manage vaccination appointments and logistics using Python.
6. Vaccination inventory system (C++): Created a C++ system for managing vaccination stock and logistics.
7. Coffee making Process (LabVIEW): Designed a process using LabVIEW to monitor and enhance coffee bean making.

8. Automated Bus Door Control (PLC & Pneumatics): Developed an automated control system for bus doors using PLC and pneumatic circuits.
9. Automotive Part Control System (C Programming): Built a system to control various automotive components using C programming.
10. Library Lighting Design (Dialux): Designed and simulated energy-efficient lighting systems for public buildings using Dialux software.
11. Automated Parking System (Arduino): Managed and contributed to the development of an automated parking system, integrating various sensors and control mechanisms.
12. Sound Purifier (MATLAB): Designed a system using MATLAB for real-time sound purification.
13. BJT Audio Amplifier (MATLAB): Built and simulated a BJT audio amplifier circuit using MATLAB.
14. Discrete time system using z-transform: Perform simulation of Discrete Time Systems using appropriate Signal Transforms.
15. Temperature Control System (Arduino & Tinker cad): Developed a temperature control system using Arduino and Tinker cad for simulation.
16. AI Hand Gesture Recognition (Python): Trained an AI model using OpenCV and Python to recognize and interpret hand gestures for various applications.
17. Develop a vision base inspection system for Fruits (PyCharm): Trained an AI model using OpenCV, Roboflow, Python to recognize Fruits.
18. Single phase transformer and three phase alternator simulation.
19. Frequency modulation & demodulation using LabVIEW.
20. Microgrid Technology Simulation (MATLAB & LabVIEW): Investigated and simulated microgrid substations using conventional and renewable energy.
21. Design of the Off-Grid Wind Energy System (Simulink): designed Off-Grid Wind Energy System that should be able to provide power for both DC and AC applications.

22. Go-Kart Design (SolidWorks): Designed a Go-Kart model using SolidWorks.
23. Hospital Logistics AGV Vision System: Developed the vision subsystem for a smart AGV using NVIDIA Jetson Orin Nano, ZED 2i stereo camera, YOLOv8, Roboflow datasets, OpenCV, PyTorch, ONNX Runtime GPU and PyQt5 to detect obstacles and medicines in real-time.
24. Medication Scanner and Obstacle Detection GUI: Built laptop and Jetson testing GUIs with camera input, confidence sliders, mask visualization, focused detection mode and medicine inventory logging to validate the trained AGV vision models before deployment.
25. Overcurrent Relay Coordination GUI (MATLAB): Developed a MATLAB App Designer tool for IDMT relay coordination, relay operating time calculation, Time-Current Characteristic curves, system analysis and validation report for ten overcurrent relays.
26. Power Flow Analysis Using MATLAB and PowerWorld: Built and analysed a 10-bus power system using Newton-Raphson, Gauss-Seidel and Fast Decoupled methods, then studied the effect of adding extra loads on bus voltage, line flow and system losses.
27. AC Microgrid Overcurrent Protection Simulation (Simulink): Built a 3-bus AC microgrid with transformer, distribution lines, CTs, circuit breakers and overcurrent relays to test IEC and IEEE inverse relay characteristics under different fault types.